

MOHAMED AMINE BEN AFIA

Data Engineer | Python Developer

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Data Engineer with 4+ years working on ETL pipelines and cloud infrastructure. Proven track record of leading migrations, mentoring engineers, and delivering measurable impact — including +40% data processing performance and -50% production incidents. Trusted on critical systems with high autonomy, rigorous quality standards, and a collaborative mindset.

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, Pandas, SQLAlchemy, FastAPI, React

Data Engineering: Apache Airflow (MWAA), Celery, ETL pipeline design, NLP

Cloud & Infrastructure: AWS (S3, Lambda, RDS, ECS/Fargate, API Gateway, CloudWatch, Secrets Manager)

DevOps & Monitoring: Docker, GitLab CI, Bitbucket Pipelines, Datadog, Pytest

Databases: MySQL, PostgreSQL

Project & Collaboration: Jira, Git, Bitbucket

PROFESSIONAL EXPERIENCE

Riseup: LMS, LXP Platform - Data Engineer

December 2021- February 2026

Leading French EdTech platform serving hundreds of schools and companies, managing multi-tenant data warehouses with several million rows and daily events.

- Led the research phase and decision-making process for the new orchestration solution (meetings with Astronomer, cost/performance comparison, final choice of MWAA)
- Led full migration of **150+** legacy Celery ETL tasks to Apache Airflow (MWAA), including environment setup, DAG development, and credential centralization via AWS Secrets Manager
- Developed multi-tenant data backend schemas and Python data layers across legacy and Airflow-based pipelines
- Achieved **+40%** data processing performance improvement and **-50%** reduction in production incidents through testing, quality gates, and end-to-end monitoring (Datadog + CloudWatch)
- Designed and deployed an MCP (Model Context Protocol) server
- Operated AWS S3, Lambda, and CloudWatch, reducing storage costs via file cleanup and compression
- Built CI/CD pipelines (Bitbucket Pipelines → GitLab CI) and a Slack bot for real-time DAG success/failure notifications
- Monitored and maintained legacy cron jobs; updated parameters and managed ad-hoc runs as needed
- Mentored and supported 4 data & backend engineers on technical topics and daily challenges
- Code review, architecture review and enforcement of engineering standards
- Managed Jira backlog, sprint planning, and task prioritization; conducted technical interviews and intern onboarding

Key Stack: Python, Pandas, Apache Airflow, AWS, MySQL, SQLAlchemy, Pytest, Datadog, Docker, GitLab CI

Rounddesk – Data Engineering & ML Internship

March 2021- June 2021

Large-scale processing of audio/video data for transcription and analytics

- Developed ETL pipelines to process large-scale audio and video data
- Integration of Google Speech-to-Text and IBM Watson APIs
- Built NLP pipeline for automatic topic detection and metadata enrichment

Key Stack: Python, Pandas, AWS S3, PostgreSQL, Google STT, IBM Watson

Carthage Solutions – Data Engineering & Data Science Internship

August 2020 - October 2020

Recommendation and skills analytics platform

- Developed ETL pipelines and content-based recommendation engine
- Analytical dashboard development (React + FastAPI)

Key Stack: Python, MySQL, React, FastAPI, Git

EDUCATION

Bachelor of Engineering in Computer Science

2018-2021

Private Higher School of Engineering of Monastir - ESPRIMS

Bachelor's Degree in Electronics, Electrical Engineering and Automation

2012-2016

Faculty of Sciences of Monastir - FSM

High School Diploma, Mathematics

2012

Khneiss High School, Monastir, Tunisia

LANGUAGES

French . English . Arabic